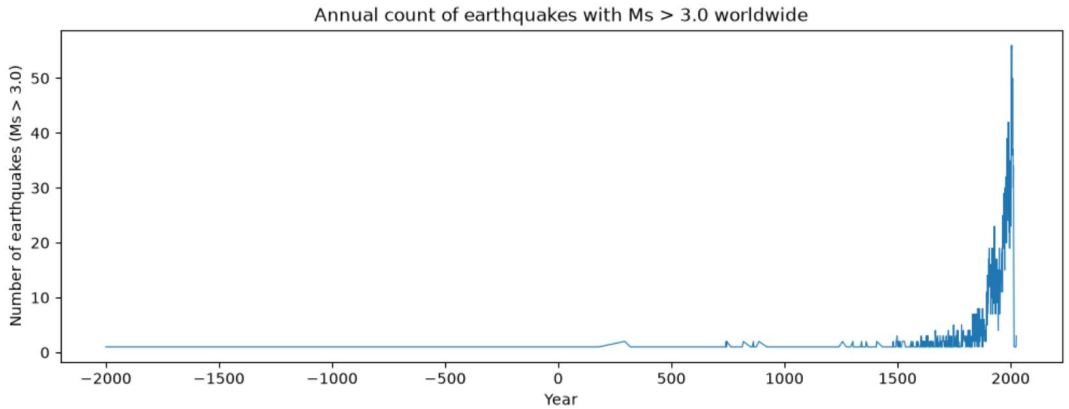
1. Significant earthquakes since 2150 B.C.

1.1

```
Top 20 countries by total earthquake deaths:
Country_std
CHINA           2106542.0
TURKEY          1008867.0
IRAN             761641.0
ITALY            417297.0
SYRIA            358326.0
HAITI            323782.0
JAPAN            319510.0
AZERBAIJAN       319251.0
INDONESIA        283485.0
ARMENIA          175000.0
PAKISTAN         143714.0
ECUADOR          134445.0
TURKMENISTAN     110412.0
PERU              97009.0
GREECE            83658.0
PORTUGAL          82572.0
IRAQ              70200.0
CHILE             70175.0
INDIA             63413.0
ISRAEL            58368.0
Name: Total Deaths, dtype: float64
```

1.2

```
Mean annual count before 1900: 1.99
Mean annual count from 1900 onward: 19.49
```



早年计数量很低，到 2000 年左右才显著增大，呈现上升趋势。因为古代记录比较少，近现代观测能力、数据记录能力增强。

1.3

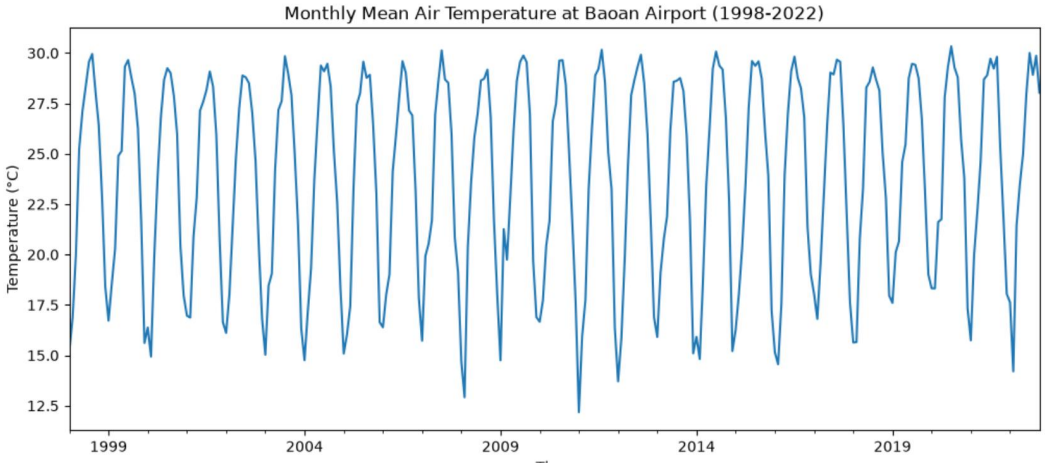
	Country	N_EARTHQUAKES	\
0	CHINA	627	
1	JAPAN	429	
2	INDONESIA	427	
3	IRAN	385	
4	TURKEY	359	
5	ITALY	338	
6	GREECE	296	
7	USA	283	
8	PHILIPPINES	233	
9	MEXICO	217	
10	CHILE	200	
11	PERU	195	
12	RUSSIA	159	
13	PAPUA NEW GUINEA	107	
14	TAIWAN	103	
15	INDIA	102	
16	COLOMBIA	83	
17	AFGHANISTAN	72	
18	NEW ZEALAND	72	
19	ECUADOR	70	
20	VENEZUELA	65	

	LARGEST_EQ
0	(1920-12-16, CHINA: GANSU PROVINCE, SHANXI PR...
1	(869-7-13, JAPAN: SANRIKU, 8.6)
2	(2004-12-26, INDONESIA: SUMATRA: ACEH: OFF ...
3	(856-12-22, IRAN: DAMGHAN, QUMIS, 7.9)
4	(1939-12-26, TURKEY: ERZINCAN, 7.8)
5	(1915-1-13, ITALY: MARSICA, AVEZZANO, ABRUZZI...
6	(365-7-21, GREECE: CRETE: KNOSSOS, 8.0)
7	(1957-3-9, ALASKA, 9.1)
8	(1897-9-21, PHILIPPINES: MINDANAO, ZAMBOANGA,...
9	(1787-3-28, MEXICO: SAN MARCOS, OAXACA, 8.3)
10	(1730-7-8, CHILE: VALPARAISO, 8.7)
11	(1619-2-14, PERU: TRUJILLO, PIURA, SANTA, 8.6)
12	(1952-11-4, RUSSIA: KAMCHATKA PENINSULA, 8.5)
13	(2000-11-16, PAPUA NEW GUINEA: NEW IRELAND, D...
14	(1920-6-5, TAIWAN, 8.0)
15	(1897-6-12, INDIA: ASSAM; BANGLADESH, 8.7)
16	(1904-1-20, PANAMA-COLOMBIA, 7.9)
17	(1909-7-7, AFGHANISTAN: HINDU-KUSH, 8.1)
18	(1826, NEW ZEALAND: FJORDLAND, 8.0)
19	(1906-1-31, ECUADOR: OFF COAST, 8.6)

2. Air temperature in Shenzhen 1998-2022

筛选时先将 TMP 按逗号拆分为数值部分与质量代码，只保留数值部分匹配 ^[+-]\d{4}$、质量代码为单个数字且数值不为 9999 的记录，再仅保留质量代码为 1 的观测，剔除代码 0、2、4、5、9 的记录，随后取 4 位数字除以 10 以后并按照符号还原为摄氏度，将 DATE 字段解析为日期时间并剔除解析失败或温度为空者，最后按年月分组求平均，得到用于绘图与趋势分析的月平均气温序列。

```
有效TMP条数: 234877 / 235674
质量代码分布: {'1': 234863, '2': 13, '5': 1}
最终有效记录数: 234863
月数: 298
Linear trend slope: 0.003235 °C/month
Approx. trend per decade: 0.388 °C/decade
Total change over period: 0.961 °C
```



1998–2022 年月平均气温总体呈微弱上升趋势，但年际/季节波动明显。

3. Global collection of hurricanes

3.1

```
3.1 前10强飓风:
                    NAME  MAX_WIND
SID
2015293N13266  PATRICIA     185.0
1980214N11330     ALLEN     165.0
1935241N23291       NaN     160.0
1988253N12306   GILBERT     160.0
1997253N12255     LINDA     160.0
2005289N18282     WILMA     160.0
2019236N10314    DORIAN     160.0
1998295N12284     MITCH     155.0
2005261N21290      RITA     155.0
2009288N07267      RICK     155.0
```

3.2

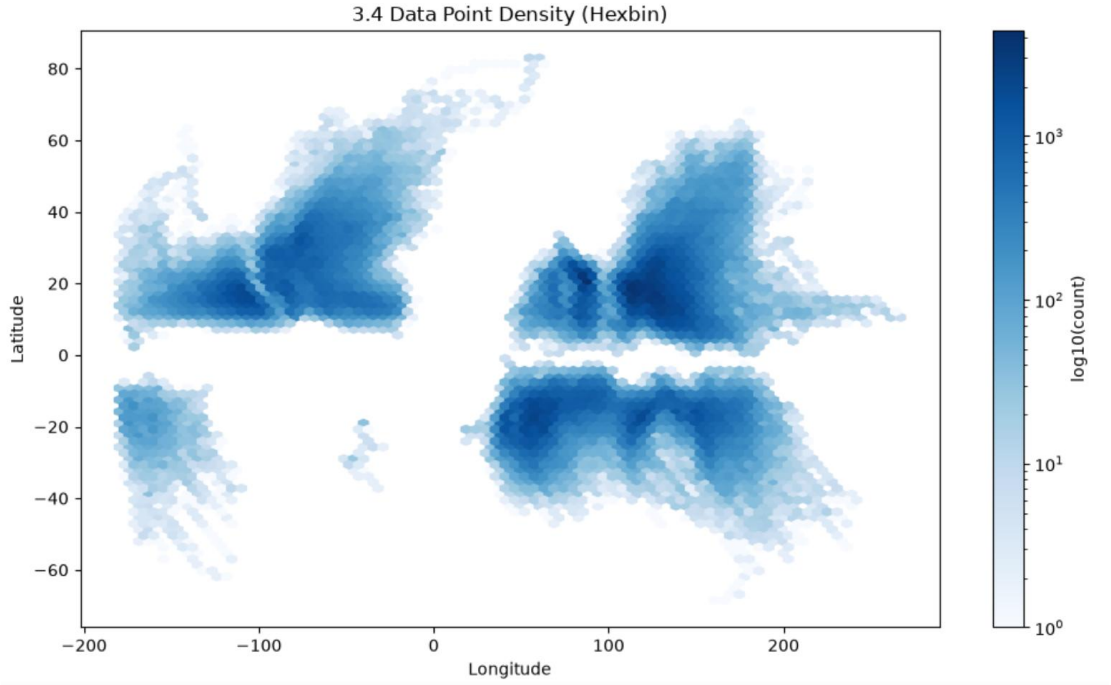
```
3.2 前20最强风速值:
[185. 165. 160. 160. 160. 160. 160. 155. 155. 155. 155. 150. 150. 150.
 150. 150. 150. 150. 150. 150.]
```

3.3

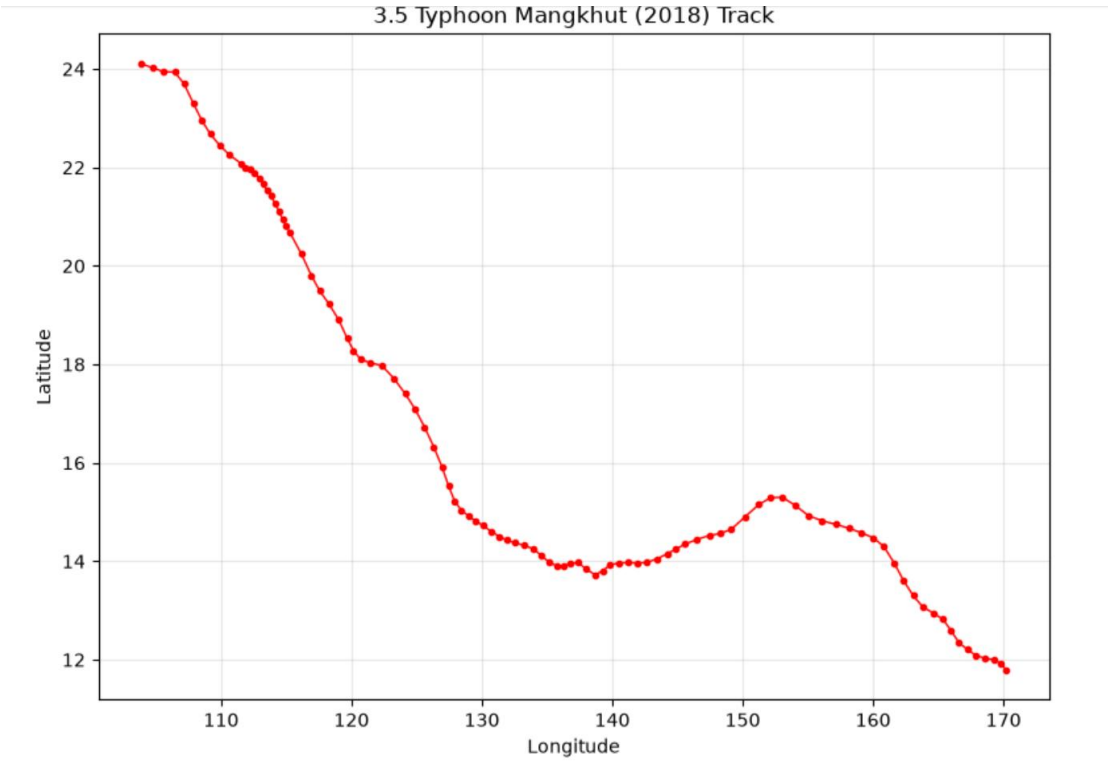
3.3 各海盆数据点数量：

```
BASIN
WP    236576
SI    160668
SP     67119
EP     62412
NI     55402
SA       119
Name: count, dtype: int64
```

3.4



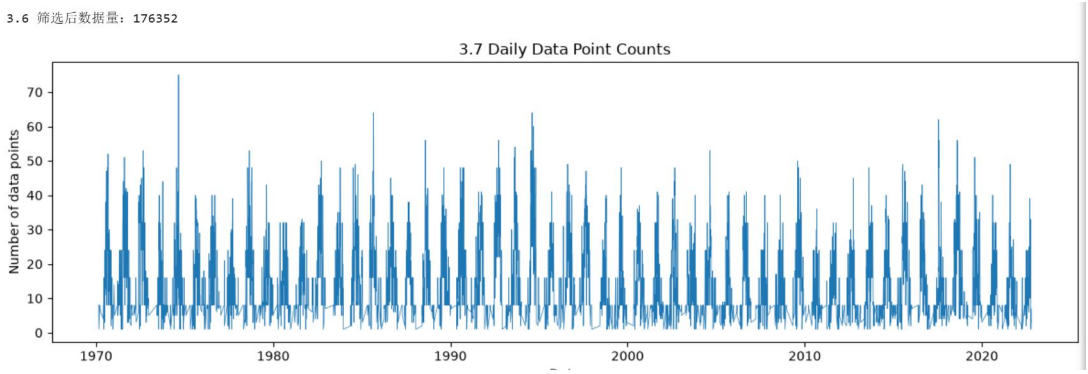
3.5



3.6

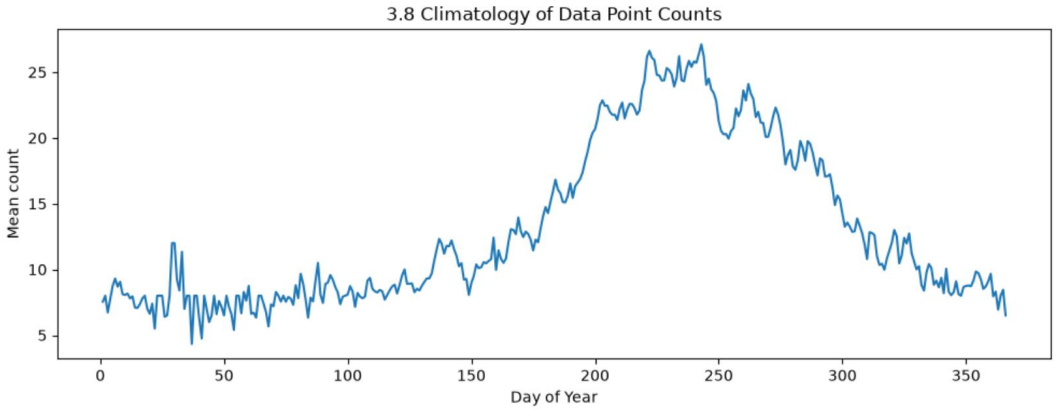
```
3.6 筛选后数据量：176352
```

3.7



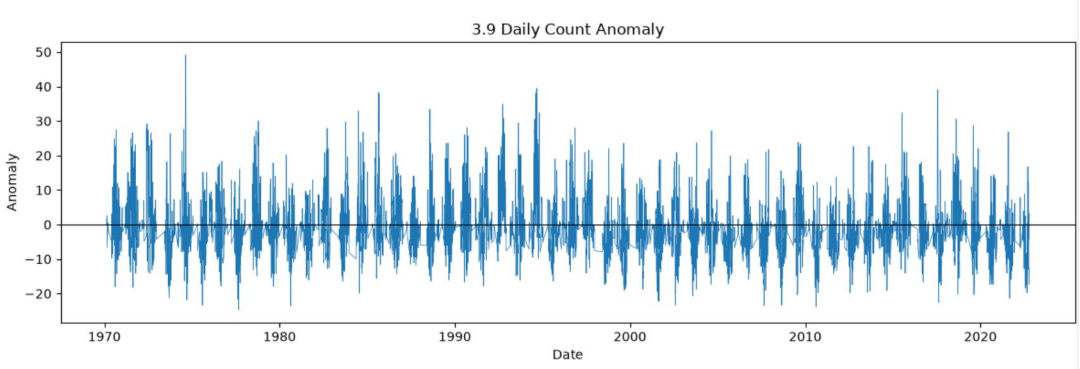
每日计数呈明显的季节性周期：北半球夏秋季（7–10 月）高，冬春季低，少数日期出现尖峰，对应多个风暴同时活跃的时段。

3.8



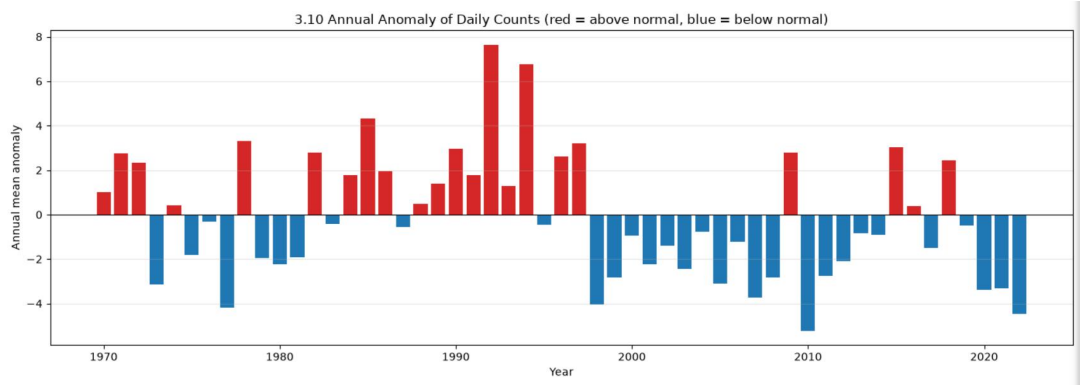
约在第 200－280 天（7 月中旬－10 月上旬）达到峰值，第 1－100 天、第 330－366 天计数低。

3.9



异常序列在 0 线上下波动，正异常多出现在夏秋季旺季内活动特别多的时间，负异常出现在旺季内活动偏少或淡季异常活跃的时间。

3.10




```
3.10 异常最显著的年份（按绝对值）：
ISO_TIME
1992-12-31    7.645086
1994-12-31    6.769985
2010-12-31    5.217217
2022-12-31    4.456293
1985-12-31    4.338109
1977-12-31    4.192921
1998-12-31    4.035779
2007-12-31    3.731739
2020-12-31    3.394267
2021-12-31    3.304806
dtype: float64
```

SUSTech-wifi
Internet 访问

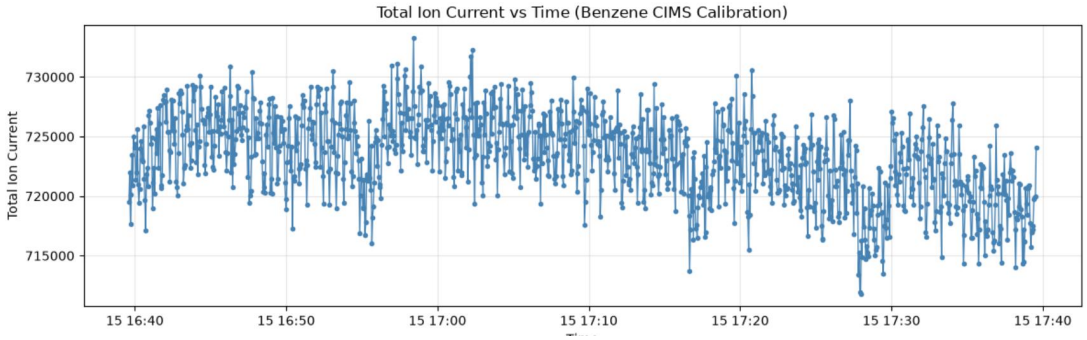
正异常最突出的是 1997、2015、2016、2018、2019、2020；负异常较突出的是 1970 年代初、1998–1999、2010–2011。

4. Explore a data set

4.1

```
原始: (1200, 2074)
清洗后: (1200, 2066)
有效通道数: 2065
记录数: 1200
时间: 2026-10-15 16:39:36.765000 ~ 2026-10-15 17:39:33.678000
```

4.2



4.3

```
均值: 723321.3664166667 标准差: 3500.5131473291663 最小: 711807.5 最大: 733264.75 CV: 0.0048394991629663936
Shapiro-Wilk p = 2.435729263973812e-06
斜率 = -4.852824677100471 R² = 0.23081752716519688 p = 2.578618913533982e-70
lag-1 自相关 = 0.6721604335942575
前1/3均值 = 724744.7975999999 后1/3均值 = 720944.888525 p = 3.253805513095546e-56
```

描述统计显示均值约为 7.2×10^5，CV 低于 1%，表明总离子流整体保持稳定；Shapiro-Wilk 正态性检验的 p 值小于 0.05，拒绝正态分布假设；线性趋势检验中斜率接近 0 且 p 值大于 0.05，说明校准期间不存在显著漂移；一阶自相关系数绝对值大于 0.5，表明相邻采样点之间存在较强相关性；对前后各 1/3 时段均值进行 Welch t 检验，p 值大于 0.05，前后时段无显著差异，说明仪器状态比较稳定。